Supporting Information

Conformational Flexibility Influences Degree of Hydration of Nucleic Acid Hybrids

Smritimoy Pramanik,[1] Satoru Nagatoishi,[1] Sarika Saxena,[1] Jhimli Bhattacharyya[1] and Naoki Sugimoto[1,2,*]

[1]Frontier Institute for Biomolecular Engineering Research (FIBER) and [2]Faculty of Frontiers of Innovative Research in Science and Technology (FIRST), Konan University, 7-1-20 Minatojima-minamimachi, Chuo-ku, Kobe 650-0047, Japan

E-mail: sugimoto@konan-u.ac.jp

Page 1: Contents.

Page 2: Table S1 for the thermodynamic parameters for the structural formations of DNA duplex.

Page 3: Table S2 for the thermodynamic parameters for the structural formations of RNA duplex.

Page 4: Table S3 for the osmolality and ln a_w values in the absence or in the presence of different osmolytes

Page 5: Figure S1 for the normalized UV melting curves of the DNA/RNA hybrid in the presence of osmolytes.

Page 6: Figure S2 for the normalized UV melting curves of the RNA/DNA hybrid in the presence of osmolytes.

Page 7: Figure S3 for the normalized UV melting curves of the DNA duplex in the presence of osmolytes.

Page 8: Figure S4 for the normalized UV melting curves of the RNA duplex in the presence of osmolytes.

Page 9: Figure S5 for the ln K_{obs} versus ln a_w of DNA and RNA duplex formations and Figure S6 for the hydration at 0 and 37°C.

Page 10: Figure S7 for the Molecular dynamics simulations for the nucleic acid structures and Figure S8 for the RMSD of the nucleic acid structures.

Table S1. Thermodynamic parameters for the structural formations of DNA duplex*

Osmolyte (wt %)	$\Delta H°$ (kcal mol^{-1})	$T\Delta S°$ (kcal mol^{-1})	$\Delta G°_{25}$ (kcal mol^{-1})	T_m (°C)
None	−52.4 ± 3.0	−41.8 ± 2.5	−10.6 ± 0.2	41.5
10% PEG 200	−48.7 ± 3.1	−38.7 ± 2.9	−10.0 ± 0.1	38.5
20% PEG 200	−46.6 ± 1.9	−37.4 ± 1.6	−9.2 ± 0.1	34.1
30% PEG 200	−44.8 ± 1.3	−36.3 ± 0.7	−8.5 ± 0.3	28.2
40% PEG 200	−41.9 ± 3.2	−34.3 ± 2.7	−7.6 ± 0.2	20.7
10% Proline	−48.2 ± 1.1	−38.3 ± 0.9	−9.9 ± 0.4	35.6
20% Proline	−47.0 ± 2.2	−37.9 ± 1.8	−9.1 ± 0.2	34.0
30% Proline	−44.2 ± 1.1	−36.0 ± 1.0	−8.2 ± 0.1	27.1
40% Proline	−40.4 ± 3.8	−33.3 ± 3.2	−7.1 ± 0.1	22.0
10% Trimethylglycine	−50.0 ± 2.8	−39.7 ± 2.4	−10.3 ± 0.1	40.6
20% Trimethylglycine	−48.5 ± 1.3	−38.6 ± 1.1	−9.9 ± 0.2	39.6
30% Trimethylglycine	−47.9 ± 3.1	−38.4 ± 2.3	−9.5 ± 0.2	35.1
40% Trimethylglycine	−46.6 ± 2.7	−37.5 ± 1.9	−9.1 ± 0.3	31.5

*All experiments were carried out in 10 mM sodium phosphate (pH 7.0), 1 M NaCl, and 1 mM Na_2EDTA, and various concentrations of osmolytes. Values are mean ± standard deviation from the curve fitting and T_m^{-1} vs. $\ln(C_t)$ plots. Melting temperature was measured at a total strand concentration of 5 μM.

Table S2. Thermodynamic parameters for the structural formations of RNA duplex*

Osmolyte (wt %)	$\Delta H°$ (kcal mol^{-1})	$T\Delta S°$ (kcal mol^{-1})	$\Delta G°_{25}$ (kcal mol^{-1})	T_m (°C)
None	−79.6 ± 1.5	−64.1 ± 2.0	−15.5 ± 0.1	55.2
10% PEG 200	−75.4 ± 1.0	−60.7 ± 0.9	−14.7 ± 0.1	52.9
20% PEG 200	−69.5 ± 3.8	−56.1 ± 3.1	−13.4 ± 0.3	49.6
30% PEG 200	−64.8 ± 0.5	−52.2 ± 0.4	−12.6 ± 0.2	45.9
40% PEG 200	−58.9 ± 1.6	−47.5 ± 0.9	−11.4 ± 0.2	41.1
10% Proline	−76.8 ± 1.4	−62.3 ± 2.3	−14.5 ± 0.2	52.3
20% Proline	−72.3 ± 2.4	−58.8 ± 1.9	−13.5 ± 0.4	48.0
30% Proline	−65.0 ± 1.8	−53.2 ± 2.8	−11.8 ± 0.5	43.1
40% Proline	−60.5 ± 2.7	−49.8 ± 3.2	−10.7 ± 0.3	38.5
10% Trimethylglycine	−77.1 ± 2.0	−62.2 ± 2.1	−14.9 ± 0.2	52.8
20% Trimethylglycine	−75.2 ± 2.4	−61.0 ± 2.3	−14.2 ± 0.2	51.7
30% Trimethylglycine	−73.2 ± 2.2	−59.6 ± 3.7	−13.6 ± 0.1	49.4
40% Trimethylglycine	−69.4 ± 1.5	−56.6 ± 2.5	−12.8 ± 0.1	46.9

* All experiments were carried out in 10 mM sodium phosphate (pH 7.0), 1 M NaCl, and 1 mM Na_2EDTA, and various concentrations of osmolytes. Values are mean ± standard deviation from the curve fitting and T_m^{-1} vs. $\ln(C_t)$ plots. Melting temperature was measured at a total strand concentration of 5 μM.

Table S3. Osmolality and ln a_w values in the absence or in the presence of different osmolytes*

Osmolyte (wt %)	Osmolality (mmol/kg)	ln a_w ($\times 10^{-2}$)
None	1822 ± 2	□3.31
10% PEG 200	2658 ± 5	–4.83
20% PEG 200	3474 ± 4	–6.31
30% PEG 200	4410 ± 6	–8.01
40% PEG 200	5132 ± 11	–9.32
10% Proline	2782 ± 10	–5.05
20% Proline	3752 ± 4	–6.81
30% Proline	4756 ± 6	–8.64
40% Proline	6044 ± 14	–11.0
10% Trimethylglycine	2480 ± 4	□4.50
20% Trimethylglycine	3418 ± 14	□6.21
30% Trimethylglycine	4812 ± 4	–8.74
40% Trimethylglycine	6028 ± 12	–11.0

*All experiments were carried out in a buffer containing 10 mM sodium phosphate (pH 7.0), 1 M NaCl, and 1 mM Na_2EDTA and osmolyte as specified. Values are mean ± standard deviation from at least three measurements.

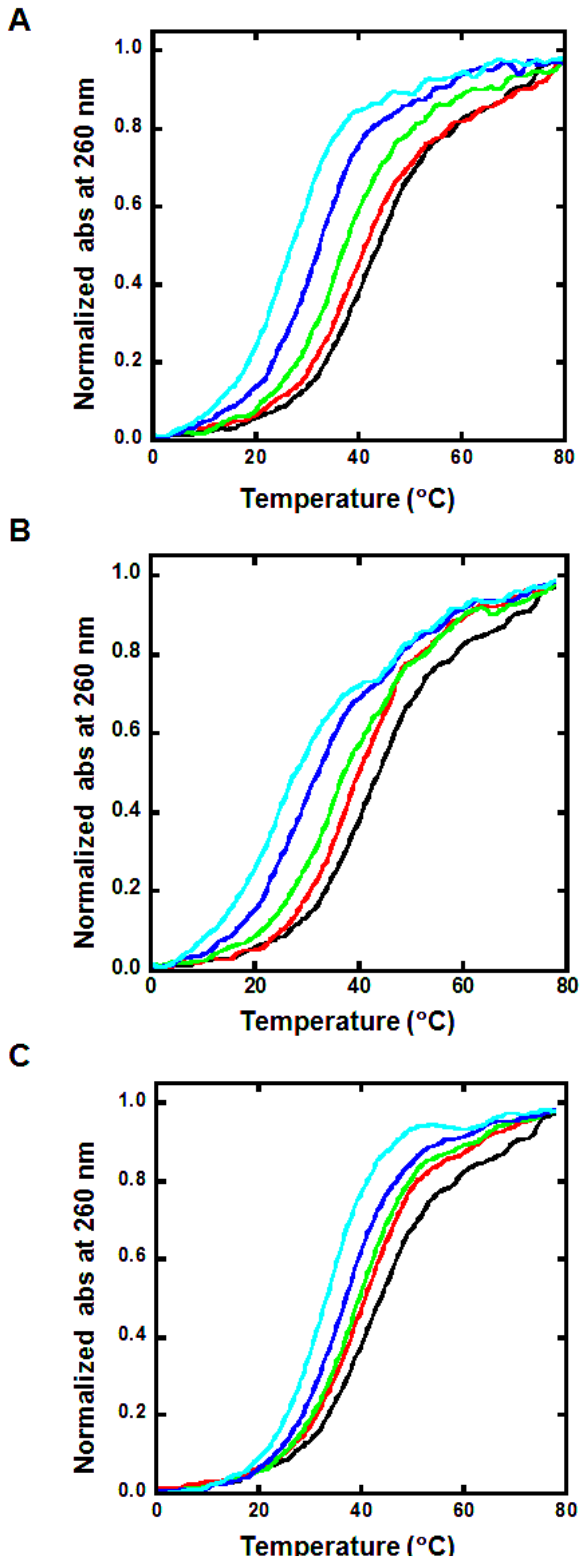

Figure. S1. Normalized UV melting curves of 5 μM DNA/RNA hybrid at 260 nm in the presence of **(A)** PEG 200, **(B)** proline, and **(C)** trimethylglycine in 10 mM sodium phosphate (pH 7.0), 1 M NaCl, and 1 mM Na_2EDTA without any osmolytes (black lines), 10 wt % osmolyte (red lines), 20 wt % osmolyte (green lines), 30 wt % osmolyte (blue lines), and 40 wt % osmolyte (cyan lines).

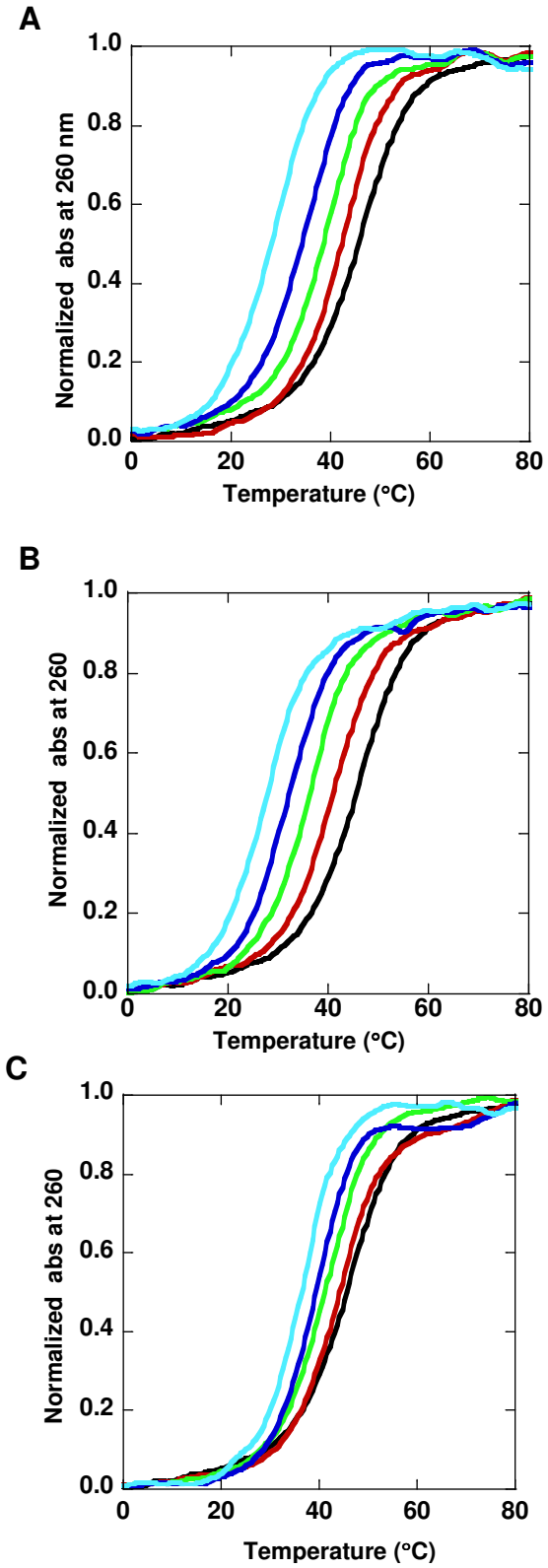

Figure S2. Normalized UV melting curves of 5 μM RNA/DNA hybrid at 260 nm in the presence of **(A)** PEG 200, **(B)** proline, and **(C)** trimethylglycine in 10 mM sodium phosphate (pH 7.0), 1 M NaCl, and 1 mM Na_2EDTA without any osmolytes (black lines), 10 wt % osmolyte (red lines), 20 wt % osmolyte (green lines), 30 wt % osmolyte (blue lines), and 40 wt % osmolyte (cyan lines).

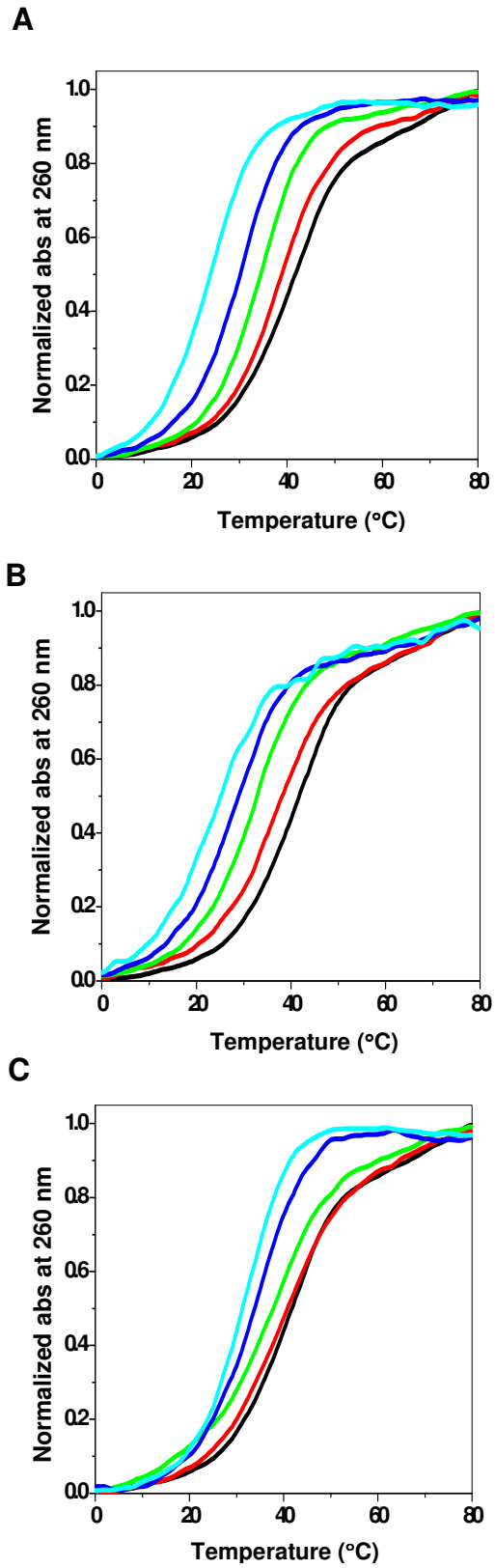

Figure S3. Normalized UV melting curves of 5 μM DNA duplex at 260 nm in the presence of **(A)** PEG 200, **(B)** proline, and **(C)** trimethylglycine in 10 mM sodium phosphate (pH 7.0), 1 M NaCl, and 1 mM Na_2EDTA without any osmolytes (black lines), 10 wt % osmolyte (red lines), 20 wt % osmolyte (green lines), 30 wt % osmolyte (blue lines), and 40 wt % osmolyte (cyan lines).

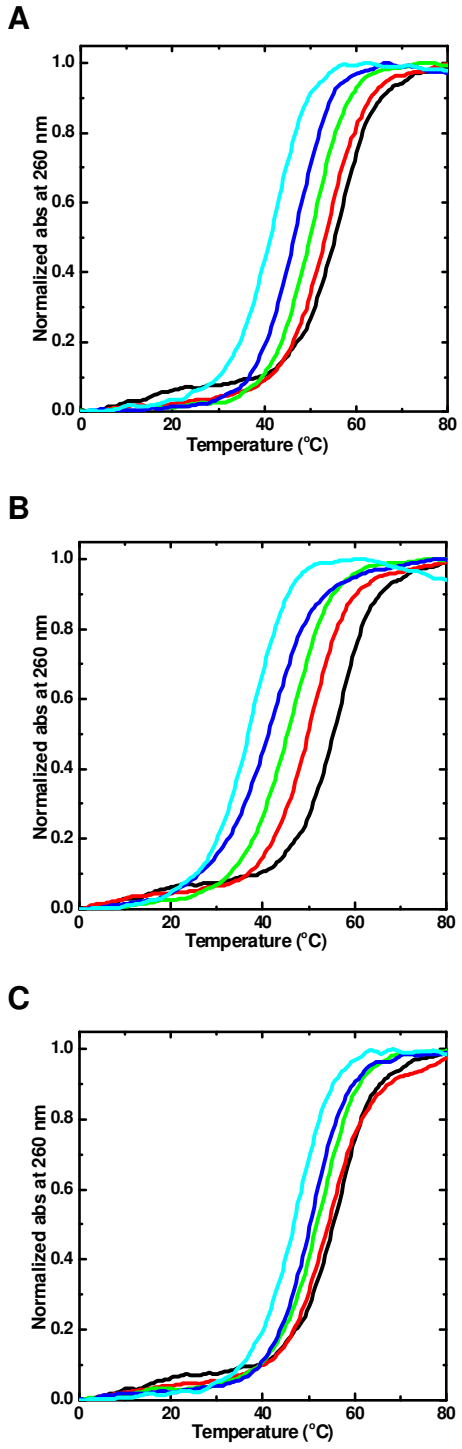

Figure S4. Normalized UV melting curves of 5 μM RNA duplex at 260 nm in the presence of **(A)**PEG 200, **(B)** proline and, **(C)** trimethylglycine in 10 mM sodium phosphate (pH 7.0), 1 M NaCl, and 1 mM Na_2EDTA, without any osmolytes (black lines), 10 wt % osmolyte (red lines), 20 wt % osmolyte (green lines), 30 wt % osmolyte (blue lines), and 40 wt % osmolyte (cyan lines).

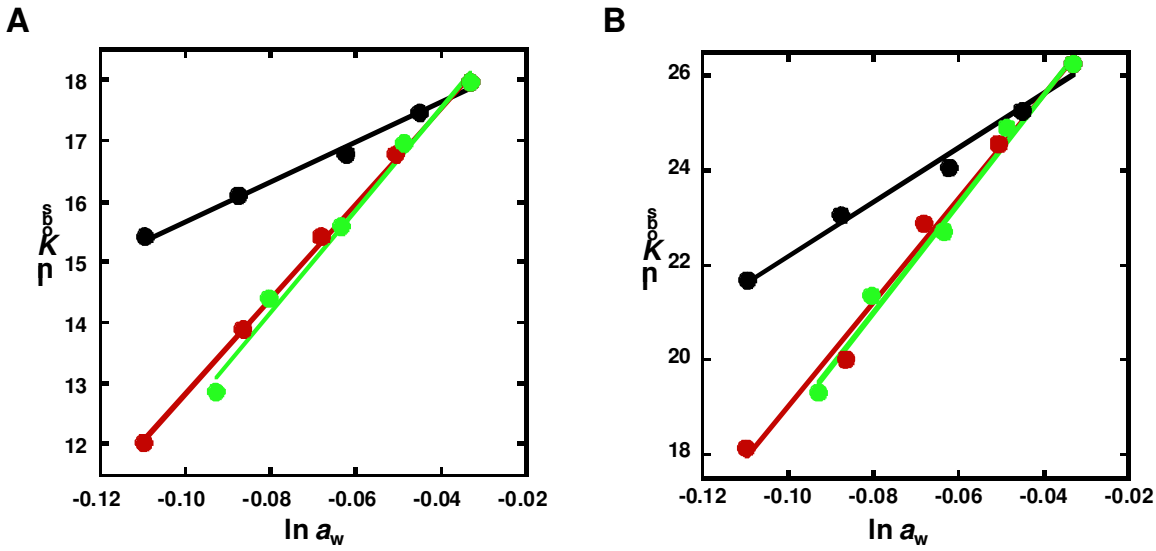

Figure S5. Plots of ln K_{obs} versus ln a_w for **(A)** DNA duplex and **(B)** RNA duplex in the presence of PEG 200 (green), proline (red), and trimethylglycine (black) in 10 mM sodium phosphate (pH 7.0), 1 M NaCl, and 1 mM Na_2EDTA with 0, 10, 20, 30 and 40 wt % osmolytes at 25 °C.

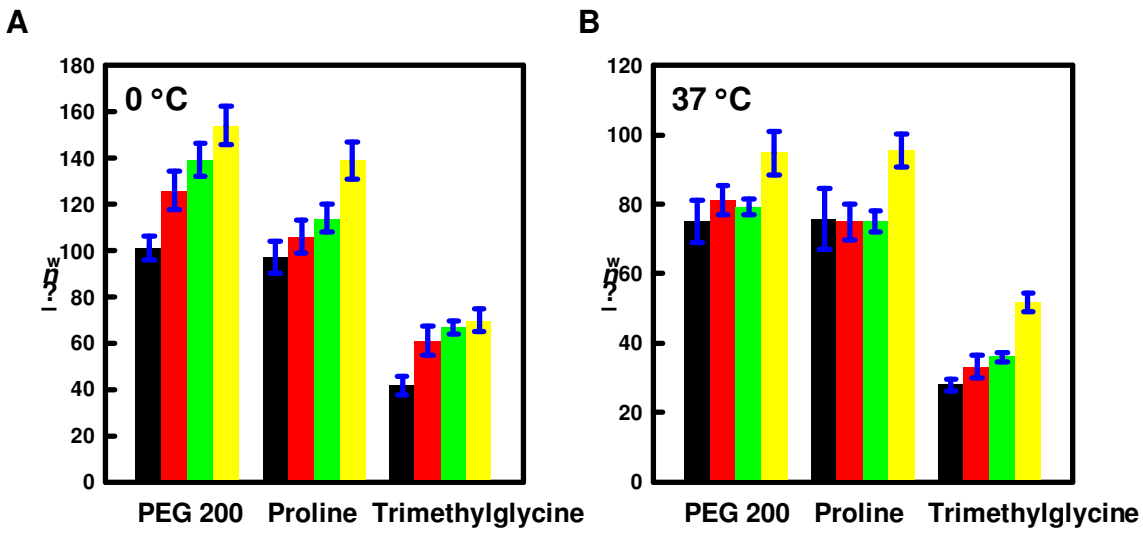

Figure S6. $-\Delta n_w$ values per duplex structures at **(A)** 0 and **(B)** 37 °C, in the presence of PEG 200, proline, and trimethylglycine for DNA duplex (black), DNA/RNA hybrid (red), RNA/DNA hybrid (green), and RNA duplex (yellow).

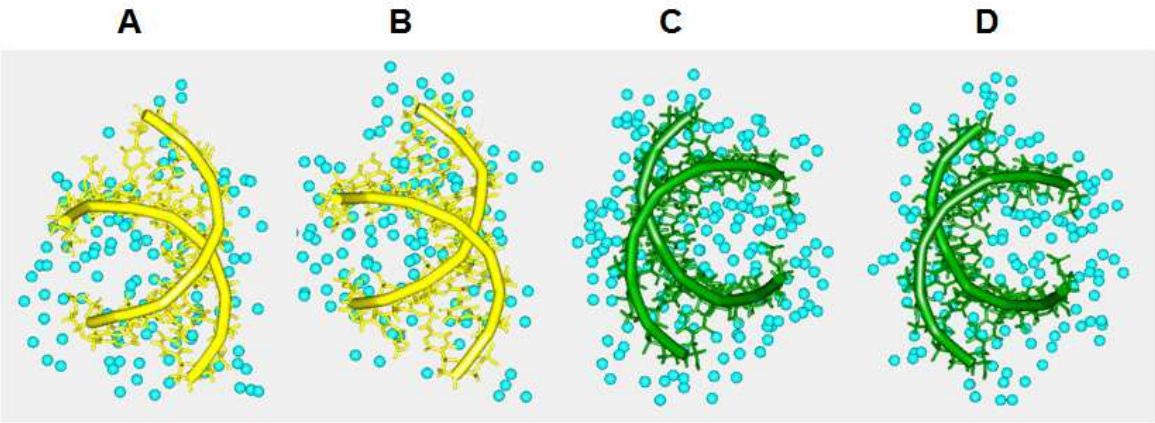

Figure S7. Snapshots of molecular dynamics simulations for **(A)** the DNA duplex at 0 °C and **(B)** the DNA duplex at 37 °C, **(C)** the RNA duplex at 0 °C, and **(D)** the RNA duplex at 37 °C, with the surrounding water molecules. DNA strand (yellow), RNA strand (green), and water (cyan).

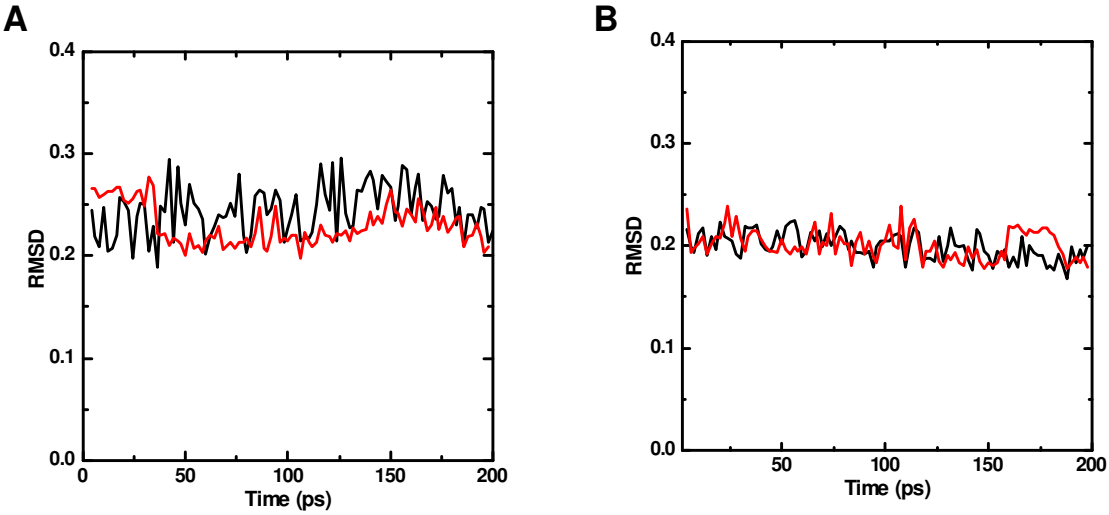

Figure S8. Time evolution of root-mean-square-displacement (RMSD) in molecular dynamics simulations for the DNA duplex (A) and the RNA duplex (B) at 0 °C (black line) and 37 °C (red line).